# Smart Modular Attachment Interface System for Kinova Jaco Robot

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**Abstract.** Upper extremity mobility impairments significantly restrict independence in activities of daily living (ADL). While assistive robotic arms like the Kinova Jaco provide support, users remain dependent on caregivers to manually switch task-specific tools due to the high precision required for standard interfaces. This paper presents the design and evaluation of a passive, smart modular attachment interface that utilizes magnetic alignment and repulsion model to enable caregiver-independent tool-switching. We compare two locking strategies: a purely magnetic "Snap-on" interface and a mechanical "Insert & Twist" bayonet latch. A user study ($N = 16$) was conducted to evaluate task efficiency and usability. Results indicate that the Snap-on system achieved significantly higher usability (SUS = 76.1 vs 61.3, $p = 0.044$) and reduced task completion time (155s vs 229s, $p = 0.010$) compared to the mechanical latch. While the mechanical latch offers higher payload capacity (6kg), the smart magnetic alignment effectively mitigates spatial and rotational ambiguity, facilitating autonomous switching.

**Keywords:** Assistive Robotics · Kinova Jaco· Modular End-Effector · Wheelchair Autonomy · Magnetic Interface

## 1 Introduction

Disorders affecting upper extremity mobility, such as cerebral palsy, tetraplegia, and spinal cord injuries, significantly restrict individuals' physical, occupational, and civic participation [12,4,13]. These restrictions lead to profound physical and social challenges, often necessitating life-long physical care [20], which impacts the quality of life (QoL) for both the users and their caregivers [14]. Regaining even partial arm and hand function is frequently cited as a primarily unmet need for improving QoL in these populations [1].

Assistive technology (AT) has the potential to increase patient autonomy, fostering better autonomy, emotional health, life satisfaction and self-esteem [16,2,18]. Particularly, assistive robotic arms integrated into wheelchairs have proven effective in helping users regain independence in Activities of Daily Living (ADL) [4,12]. However, clinical evaluations show that the functional impact

of these robots remains limited for specific tasks; for instance, written communication or fine grooming remains difficult with standard factory-shipped grippers [4,5]. Furthermore, while users feel empowered by the technology, they often remain dependent on caregivers to set up the robot or swap tools for different tasks, meaning the actual burden on caregivers is not always reduced [12,4].

Refocusing robotic assistance toward task-specialized grippers, rather than relying on simple pick-and-place, has been identified as a critical path forward [5,12]. While automated tool-switching systems are common in industrial manufacturing [7], they rely on heavy pneumatic or electrical actuation [11,19]. These active systems are fundamentally incompatible with the low-payload capacity ($\approx$ 2.6 kg) and battery constraints of consumer-grade assistive robots like the Kinova Jaco [10]. Although some commercial and open-source modular interfaces exist to allow for custom tool attachments [9], they still require a caregiver to manually perform the switch. To our knowledge, no existing system allows for completely passive, safe, and independent tool swapping by the user.

The central aim of this study is to extend a modular end-effector interface for the Jaco robot, developed by our research group [9], into a system that enables users to switch tools independently. However, this system requires a caregiver to manually assist with every step of the tool-switching process, limiting user autonomy. Our contributions are 1) the design of two passive smart, auto-aligning devices, utilising different locking strategies: a magnetic "Snap-on" system and a mechanical "Insert & Twist" system, and 2) a comparative user evaluation ($N = 16$) assessing performance through timing and usability metrics (SUS).

This work is structured as follows: In section 2, we show the iterative mechanical design process of the two proposed passive attachment interfaces. Section 3 outlines the user evaluation methodology. Section 4 provides a critical discussion of the engineering trade-offs, and Section 5 concludes the paper and proposes directions for future work.

## 2 Mechanical Design

To develop the attachment protoypes, an iterative design-build-test cycle was employed by the authors, testing connection with only the robot. The following sections present the result of the final iteration.

### 2.1 Baseline and Redesign Requirements

We parted from the original open source design by Green and Nunez Sardinha [9] (Figure 1). The design works as two components: the Palm, a simple piece with a bayonet-style connection on one side, attached to the Jaco arm with flexible straps (Figure 1).a and b), and the modular tools, which have a complementary peg, connected by inserting and twisting (Figure 1).c). Different modular tools can be designed, allowing the Jaco arm to handle accessories such as pens, cutlery, and stamps in a stable manner, and without impeding opening or closing the end-effector.




(a) Jaco-to-Palm Connection.

(b) Palm-to-Tool Connection.

(c) Jaco Gripper with Attachment Interface.

Fig. 1: Open Source attachment interface, orinally by Green and Nunez Sardinha [9].

While stable, the modular attachments are intended to be connected by a caregiver. These contradict the principles of an “ideal” robotic locking mechanism, which should feature adjustable locking directions, short switch times, and the ability to engage without requiring the user to apply complex, multi-directional forces [15]. The design lacks centering features and requires high-precision alignment. To enable autonomous use, our redesign prioritized three requirements: (1) high tolerance for spatial misalignment via "lead-in" geometry, (2) elimination of rotational ambiguity through magnetic indexing combined with free rotation in both directions, and (3) a passive ejection mechanism.

### 2.2 Smart Magnetic Array and Orientation Logic

The core contribution is a "smart" magnetic array that functions as a passive alignment controller. The Palm is embedded with four NdFeB magnets in an alternating $\{N, S, N, S\}$ square pattern. The attachment interface has two rotation dependent modes, engage Fig.2 and disengage Fig.3 that seamlessly allow the user to pick up and drop the tool. This configuration creates a discrete-state potential field that guides the Tool into distinct rotational modes depending on the orientation:

**Total Attraction/Engagement (0° :** The magnets align to opposite poles, providing maximum attraction force to seat the tool. In the Insert & twist configuration, the latch is engaged in this configuration, constraining one DOF.

**Partial engagement (45°):** The magnetic force pulls to twists the tool to the Attraction mode, but small vertical is present. The user must approach at 45° to align the bayonet wings with the entry slots.

**Repulsion/Ejection (90°):** Rotating the Palm 90° aligns like-poles (N-N, S-S). This pushes the tool away from the Palm, ensuring a clean release into the holster. This action is only feasible while the tool is held in the holster. In the Insert & Twistthis simultaneously clears the mechanical latch.

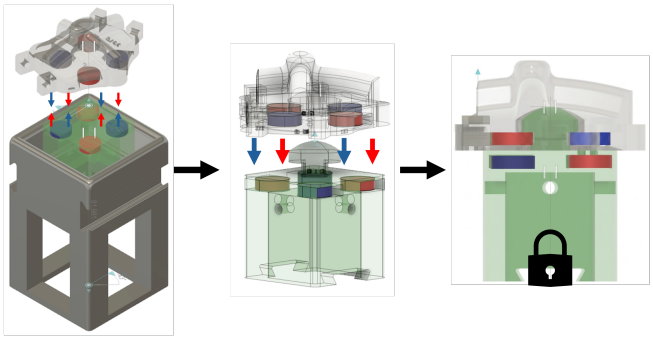


Fig. 2: Jaco-Palm approaching tool for attachment. Magnetic force aligns the magnets and holds the attachment in place.

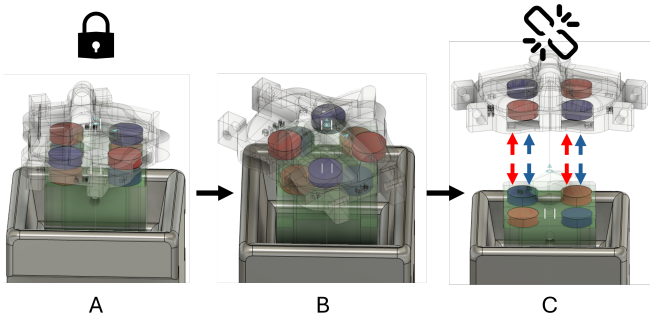



Fig. 3: Jaco-Palm twisting to drop tool after re-insertion. Magnetic force then repels the accessory, keeping it in the holster.

### 2.3 Locking mechanism

Using the magnetic base, we developed a new variation incorporating the mechanical bayonet latch from the baseline design. This intends to address the trade-off between usability and payload capacity (Fig. 4). Thus, this **Snap-on prototype** relies exclusively on magnetic flux for retention, without the need of mechanical obstructions, allowing the user to approach the tool at multiple angles.

The defining feature of the insert & twist tool prototype is a mechanical latching mechanism (Figure 4a). Unlike the magnetic retention of the snap-on interface, this system utilizes a bayonet-style locking geometry to ensure a high-integrity connection. Once the tool is fully seated and rotated into the locked position, the latching interface constrains five degrees of freedom (5-DOF), restricting all translational movement and axial separation. This mechanical interlock allows the system to support significantly higher payloads and withstand greater external forces compared to the snap-on variant.



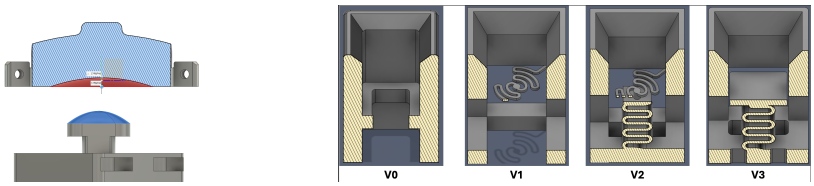
(a) Insert & Twist prototype, with mechanical latch for secure and rigid locking.

(b) Snap-on lock mechanism, relying only on magnetic attraction for tool retention

Fig. 4: Overview of attachment interface features, showing a cross section of the snapping mechanism. The piece in green is the peg, sitting behind the slot after rotation.

### 2.4 Shared Passive Guidance Features

Two features were implemented across all prototypes to facilitate independent switching:




(a) Tool end loft for self guidance

(b) Holster cross section, showing chamfer and potential compliant features.

Fig. 5: Overview of attachment self guiding features.

1. **3D Convergent Geometry:** Both the Palm and Tool ends utilize lofted, rounded "peg-and-socket" profiles (Fig. 5a). These features provide a ±5mm lateral offset tolerance, mechanically guiding the tool into the seated position using the robot's downward force.
2. **Tool Holster:** A specialized holster was developed to support the tools and to provide counter-torque during the locking/unlocking phases. It features chamfered lead-ins to facilitate the "drop-off" phase and a modular dovetail system for tool-library expansion (Fig. 5b). Compliant features can be added to support tool switching and protect the arm.

### 2.5 Mechanical Payload Capacity

Static load testing was conducted to evaluate the mechanical retention of both prototypes against the Jaco arm's 2.6 kg payload limit. The passive magnetic retention of the **Snap-on** interface was emasured with a spring balance, sustaining



Fig. 6: Experiment set up for users to trial the developed attachment interfaces

a maximum static load of approximately 350–400 grams / 3.4-3.9N before the magnetic attraction was overcome. The bayonet latch of the **Insert & Twist** was measured by adding progressive loads, supporting a 6 kg / 58.8N suspended deadweight without mechanical failure, safely exceeding the robot's rated payload capacity.

## 3 User Evaluation

To evaluate practical efficacy, a user study was conducted to compare the performance of the two interfaces.

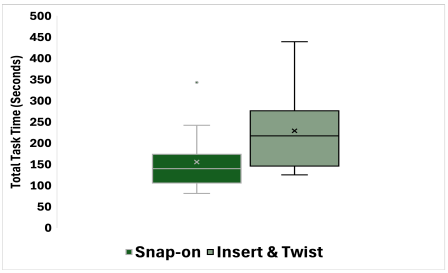
### 3.1 Testing Methodology

The study included 16 able-bodied novice participants ($M = 21.8$ years, $SD = 3.1$) in a within subjects condition. After a brief training session, each participant executed the experimental task: Utilising the Jaco with its joystick in Cartesian mode to navigate to the holster, engage the tool, utilise it to stamp a piece of paper, and safely disengage the tool back into the holster. To ensure participant safety, the experiment took place in a controlled setup with the jaco arm attached to a table, the holster on a box, and the participant on a chair. After each condition, participants filled a questionnaire. Condition order was randomised to prevent learning bias, and all tests started from the Jaco 'home' position. Performance was assessed quantitatively via phase-timing data and qualitatively via the System Usability Scale (SUS) and Likert-scale surveys.

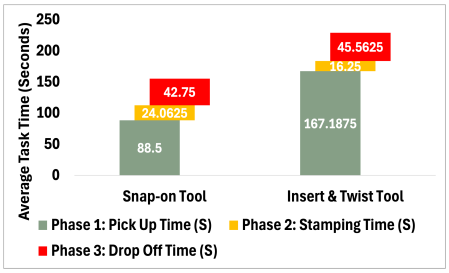
Results are presented as $MEAN(\pm STD)$. Paired sample t-tests was performed on the data collected to understand the significance of results, the threshold for significance was assumed for $p < 0.05$.

### 3.2 Results

**Task Performance Results** The magnetic Snap-on prototype performed significantly faster 155.31($\pm$67.28)secs, than the mechanical Insert & Twist prototype 229.00($\pm$95.93)secs, ($p-val = 0.010$). As illustrated in Fig. 7a, the Snap-on interface demonstrated a highly compressed interquartile range for task completion times. Conversely, the Insert & Twist mechanism exhibited extended median

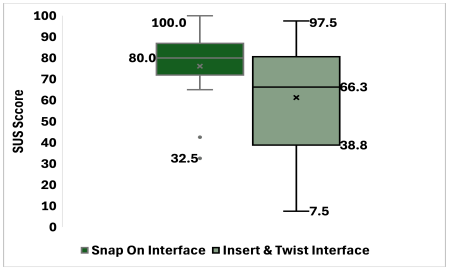



(a) Total Task Time per Tool split by gender. Outliers show delay for misalignment.

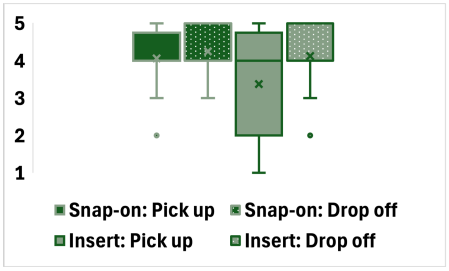



(b) Average Task Time broken down by task phase.

Fig. 7: Quantitative operational efficiency results.




(a) System Usability Scale scores distribution for both attachments.




(b) Ease of Pick-Up & Drop-Off Ratings distributions.

Fig. 8: Qualitative user perception and system usability metrics.

times and extreme upper outliers (exceeding 400 seconds). Phased timing data revealed that this delay was isolated primarily to the pick-up phase, as the bayonet latch penalised users for minor misalignments prior to rotation. Splitting by gender did not lead to significant differences (P-val=0.57).

**Usability and User Perception** The SUS evaluations further corroborated the quantitative timing data. Participants rated the overall usability of the Snap-on interface 76.1($\pm$17.25) significantly higher than the Insert & Twist interface 61.3($\pm$25.45) representing a statistically significant difference ($p - val = 0.044$). Based on standard adjective rating scales [3], the Snap-on system ranks between 'Good' and 'Excellent'.

Furthermore, Likert-scale feedback regarding the ease of specific task phases aligns closely with the timing outcomes. The Snap-on tool was perceived as noticeably easier to acquire during the pick-up phase, 4.1($\pm$0.8) vs Insert and twist 3.4($\pm$1.4) with results approaching statistical significance ($p-val = 0.052$). Conversely, there was no significant difference in user ratings for the ease of drop-off with both scoring around 4.2($\pm$0.8) ($p-val = 0.633$). This lack of significance

is expected and aligns perfectly with the mechanical design, as both prototypes utilise the same drop-off procedure and holster geometry.

## 4 Discussion & Limitations

The experimental data confirms that, using our smart magnetic attachment, carer-independent tool-switching is possible, allowing 16 naive participants to perform the switch. The additional metrics revealed a fundamental engineering trade-off between kinematic usability and mechanical payload capacity, with participants performing significantly faster, giving higher SUS scores, and borderline significant preference for the Snap-on tool during pick-up, despite its lower mechanical payload. This could be attributed to the easier process, requiring no complicated, hard-to-gauge directional forces, commonly recommended in the literature [15]. Furthermore, by allowing bayonet mount to spin 360 degrees and in either direction, the connection point eliminates rotational ambiguity that previously caused users to struggle. However, its 400g payload limit confines it to lightweight ADLs (e.g., stamping, painting). This points to a drawback, suggesting that heavier or higher power tools benefit from the twist attachment, while lightweight or frequtenly swapped ones benefit from the snap-on. The tool could potentially reduce task-load, but dedicated evaluation is needed.

While Recent developments in autonomous manipulation prioritize active electromechanical locking systems, such as 24V solenoid latches [11] or motorized hexagonal interfaces [19], the developed prototypes presented rely entirely on passive engagement, successfully bypassing heavy electrical overheads.

Although no direct comparison with our specific tasks was available in the literature, our timing results for picking up fine objects were faster— 155.31($\pm$67.28)secs with the Snap-On and 229.00($\pm$95.93)secs with the Insert & Twist — than those reported in a similar study that measured the time required to pick up and turn a key in a lock using the original assistive robot gripper, which took 278.29($\pm$270.75)secs (Study Gripper), with key lifting alone taking 204.42($\pm$173.85)secs. These findings support the need for implementing simpler pick-up strategies in assistive robots. It should also be emphasized that our participants were first-time users, whereas the other study included experienced users with disabilities. [6]

Paper [12] describes a comparable experiment in which participants carried out a set of pick-and-place and precision-intensive tasks. In contrast to their findings, every participant in our study successfully completed the task on the first attempt; however, it is important to emphasize that the papers participants were experienced powered-wheelchair users with upper-limb mobility impairments. This may indicate that having such a tool facilitates task completion, though a direct comparison is challenging without specifically controlled, dedicated testing.

The study's findings are primarily limited by its participant demographics and a highly controlled testing environment. The relatively small sample size ($N = 16$) consisted entirely of able-bodied novices aged 18–30, limiting applica-

bility to powered wheelchair users with limited upper-extremity mobility. Furthermore, the holster was mounted to a stationary box using Velcro, which fails to replicate real-world applications where the device would be rigidly mounted to a dynamic, moving wheelchair. This does not account for practical variables such as wheelchair vibration, varied approach angles, and spatial constraints. Longitudinal clinical trials with users and their caregivers in a realistic environment similar to Beaudoin *et al.* [4], are needed to accurately capture impact, along with evaluating the effect in daily caregiving time and burden.

## 5 Conclusion & Future Work

This study successfully designed and evaluated two novel attachment interfaces to enable independent, modular tool-switching for the Kinova Jaco robotic arm. Results demonstrated the utility and mechanical feasibility of the magnetic mechanism for docking. By modelling the mechanical interface to have convergent magnetic arrays, we reduce the identification, rotational ambiguity and spatial alignment requirements for the user, allowing successful docking despite the inherent noise in manual robotic teleoperation and without relying on heavy motorised locking mechanisms. Participants reported the Snap-on tool as more usable and faster to equip, but sacrificing payload capacity, suggesting that the correct connection mechanism may be task-dependant. The magnetic system can provide and accessible, economical and customizable solution to potentially increase the daily autonomy of wheelchair users and reducing caregiving hours.

The pick up drop off process can be further facilitated by automation: (a) implementing software to automate pick-up and drop-off, (b) allowing tool attachment/detachment without need for rotation, (c) Improved method for attaching tools to the tool end, i.e., how pens, stamps, and other items are mounted on the tool end-effector; at present, screws are used to hold them in place.[8,17]

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